# A modern Data Platform

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**Microsoft Denmark**

Microsoft

**Agenda / topics**

What would we like to do with our data ?

Why is cloud interesting ?

Artificial Intelligence as in AI/ML/Gen/Agentic AI

A data platform – what it could look like

**Overall theme** – compliance, governance and cost-aware

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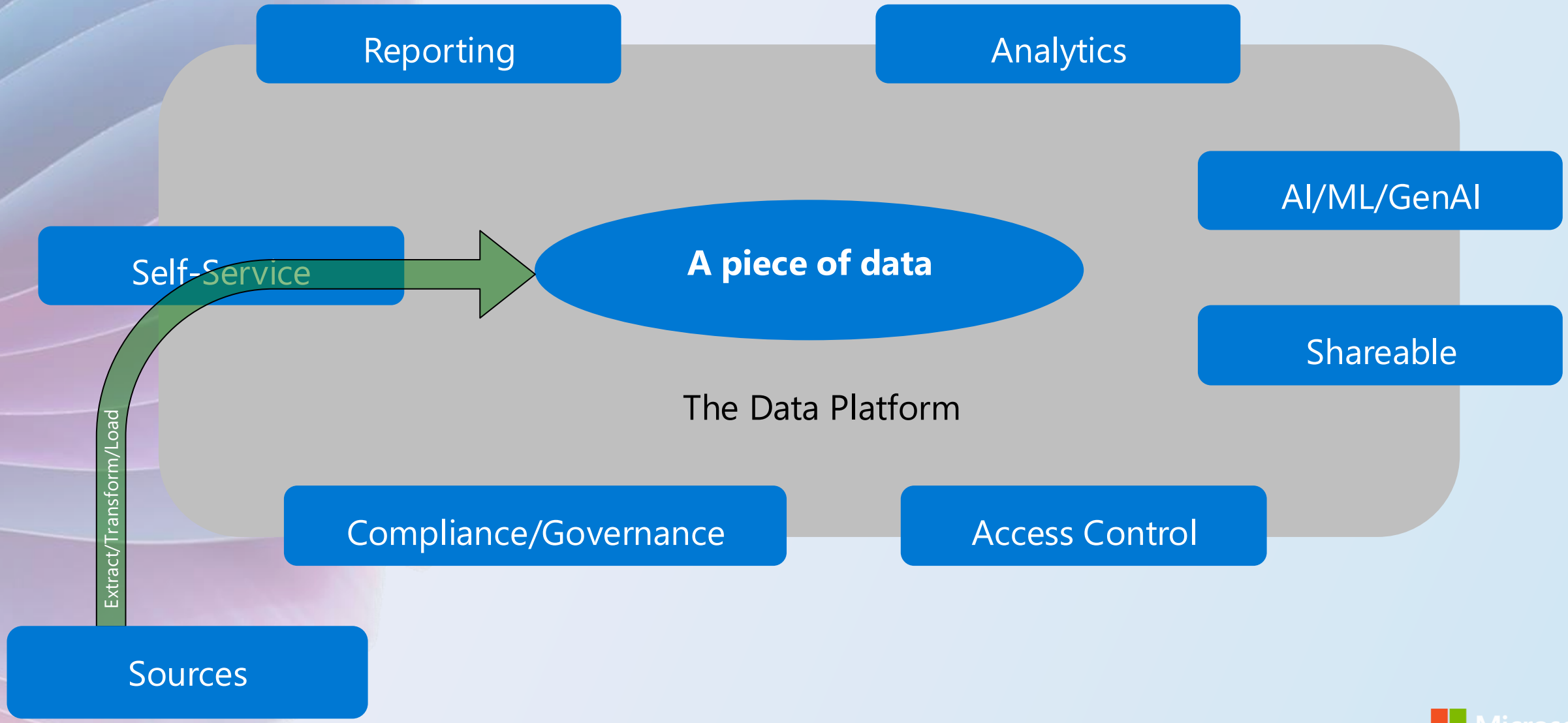
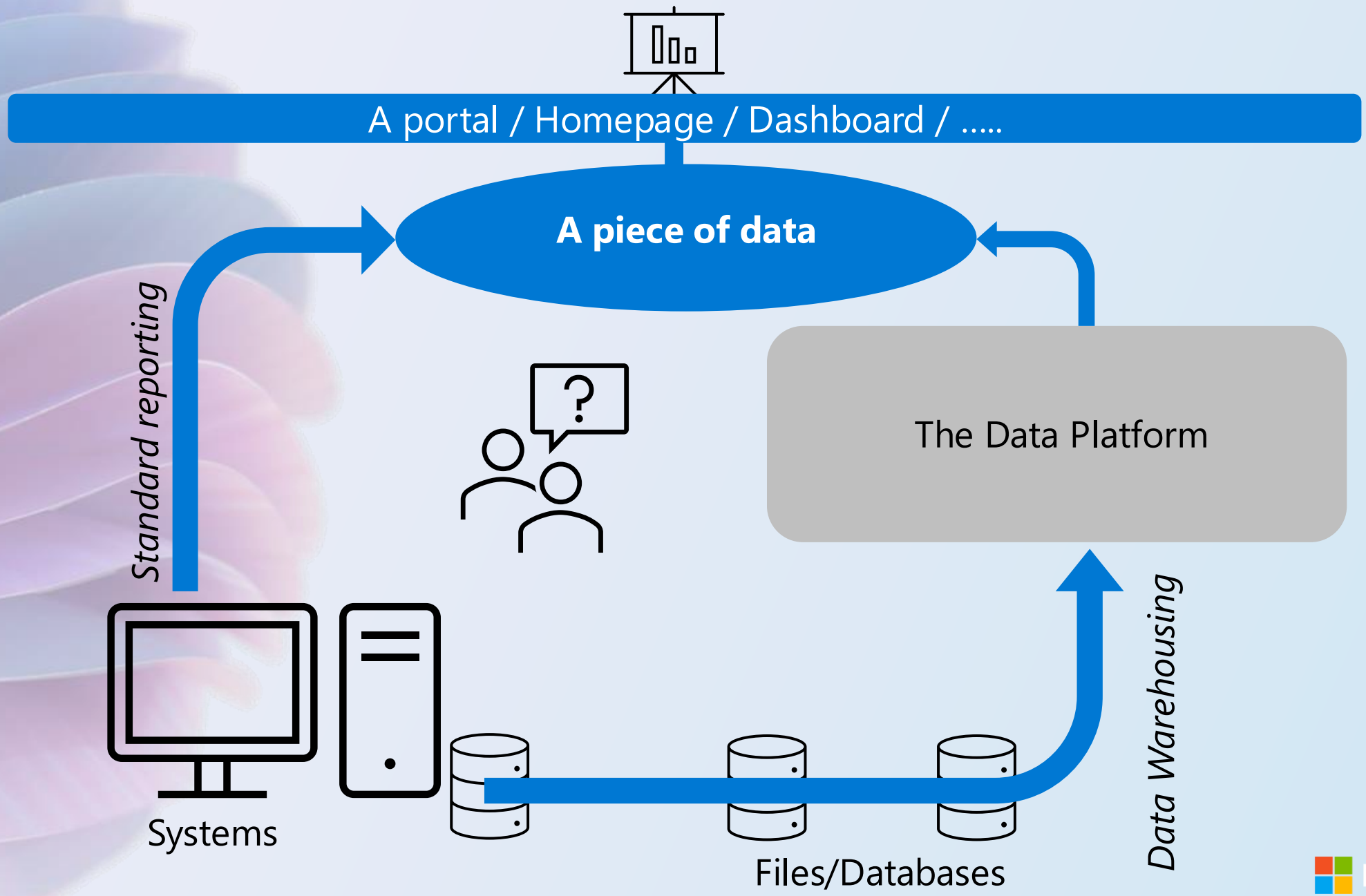
# Data – how to use them


Reporting
Analytics
AI/ML/GenAI
Self-Service
A piece of data
Shareable
The Data Platform
Extract/Transform/Load
Compliance/Governance
Access Control
Sources

Microsoft


A portal / Homepage / Dashboard / .....
A piece of data
Standard reporting
The Data Platform
Data Warehousing
Systems
Files/Databases

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# The Cloud

Azure

Microsoft

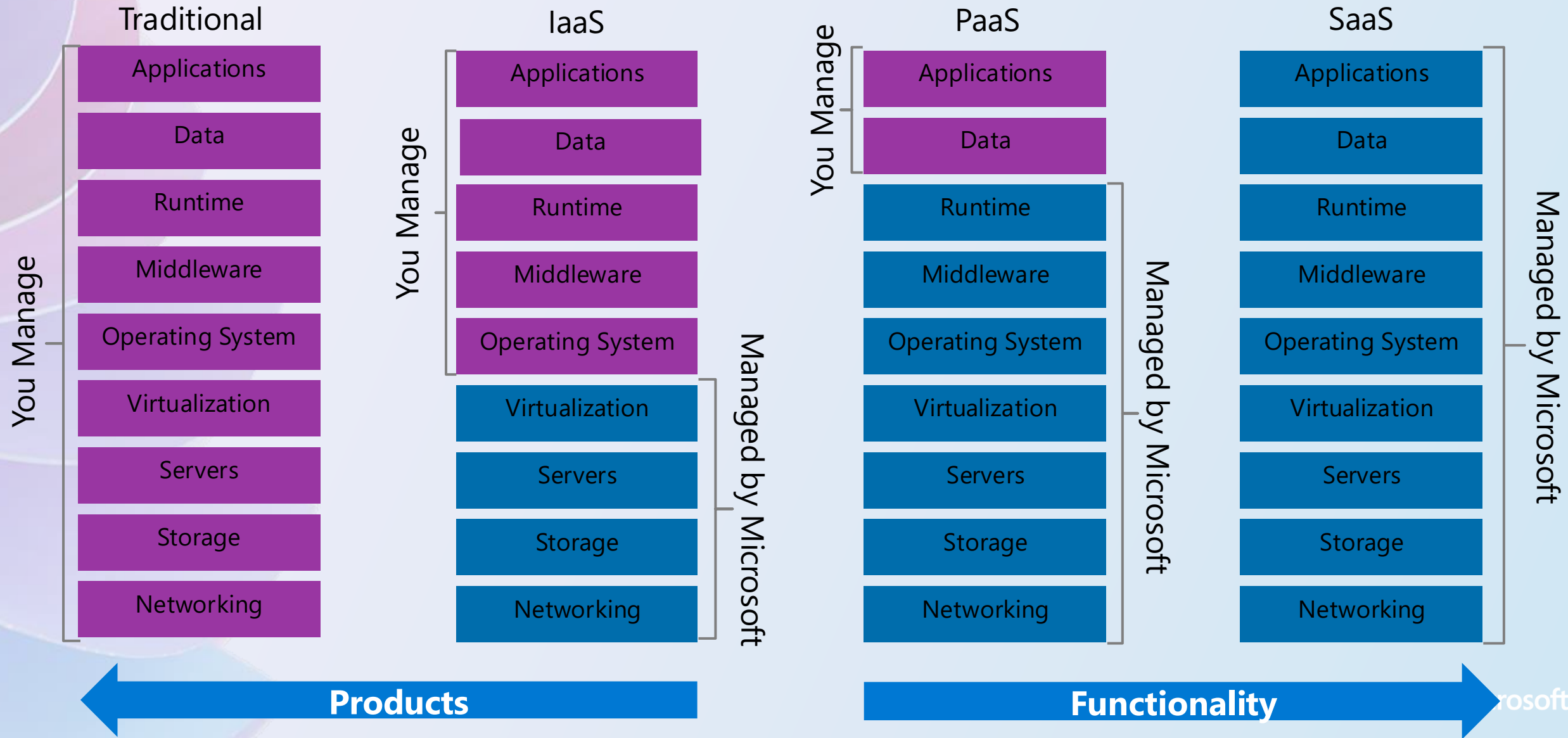
# Azure – Denmark East

- One region with three underlying physical locations at Roskilde, Køge and Høje Tåstrup
- Three locations ensure Microsoft can deliver high SLA
- Under construction and going live in 2025

**Why use Danish region?**

- Hyper-scale cloud
- Security and compliance
- Local innovation and growth
- Green IT operations ESG
- Data at Danish location
- Very low latency

# Cloud Models



# Azure runs what you need

Microsoft is one of the biggest OpenSource contributors

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# A data platform

“Future proof”

# Data : Capabilities

| Capability | Description |
|---|---|
| Consistency | A consistent solution design to ensure simple operation and further development of solutions. Replacing resources must be flexible. |
| Modulated | Solutions in the data infrastructure are developed modularly with a well-defined interface, so it is easy to replace subcomponents. |
| Technology independency | The architecture must be independent of the technology being used. This means that no matter what technology is being used, the different processes, functionalities and layers are the same. |
| Scalability | Scalability (up/down) is part of the solution design from the beginning, so that implementation and operations are not affected by bottlenecks, downtime or a possible mandatory purchase of unforeseen licenses. |
| Agility | The focus is on MVP (minimum viable product) and ongoing feedback to previous steps in the data flow.<br>It is OK to fail, and solution designs must be tested as soon as possible during the development process. |
| Security | Security is incorporated into the general architecture as well as in the concrete solution designs, both in terms of information security and privacy. |
| Data encapsulation | Data in the data infrastructure is accessed through an interface that enables control of who has access to what. The interface allows you to make changes to the data infrastructure without affecting external systems. |
| Reuse | Solutions are being developed for reuse. The architecture will consist of templates for solution designs that shorten time-to-market and ensure standardization. |
| Feedback | The architecture is continuously adapted and improved based on feedback from the different users of data. |

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Source systems

Streaming/ events

Source systems

Batch

## Ingest

Characteristics

- Consume raw data
- High degree of integration points
- Very technical
- No transformation

## Transform

Characteristics

- Technical usable data
- Getting data from ingest layer - only
- Very technical
- Place for technical transformations
- Adhere to company chosen standards

## Publish

Characteristics

- Business useable datasets
- Datasets placed in whatever suitable technology for the end usage
- “Lives” as long as being used
- Can be recreated (by self-service)

## Consume

Characteristics

- End user tools
- Business Intelligence
- Reporting
- Machine Learning
- Artificial Intelligence

Supporting Data Services

Data Catalog | Common Data Model | Data Discovery | Data Lineage | Data Governance | Master Data

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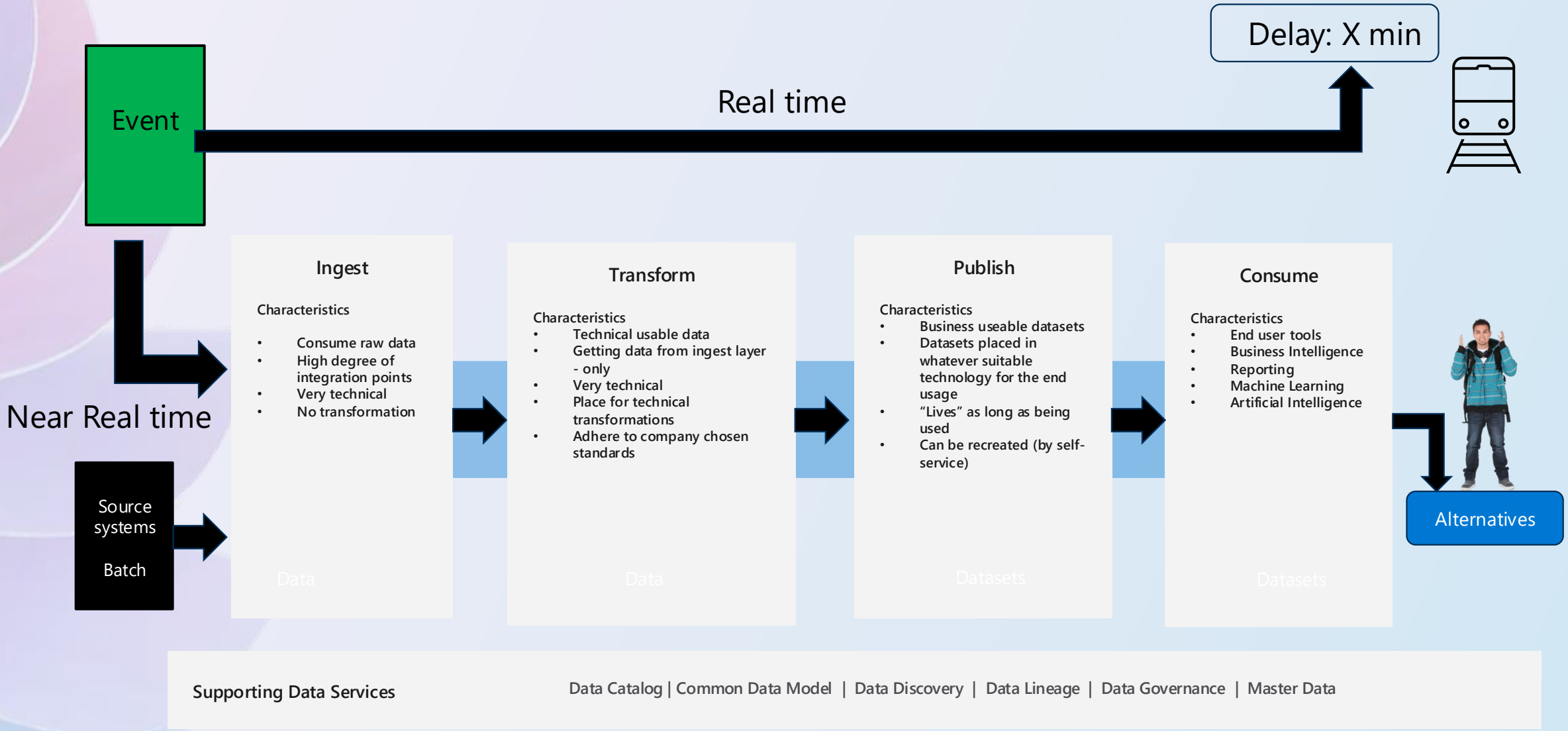


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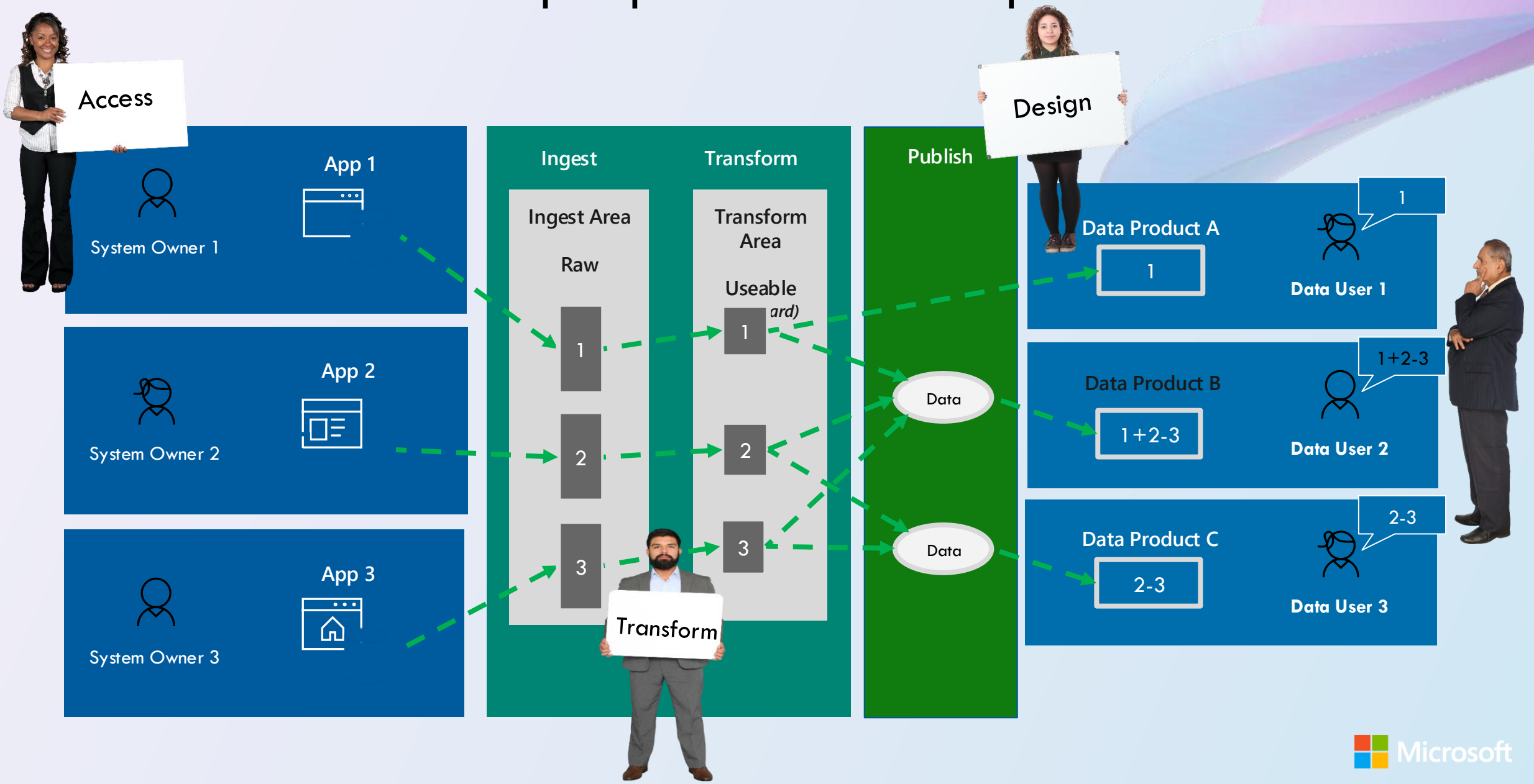
# The purpose of a dataplatform

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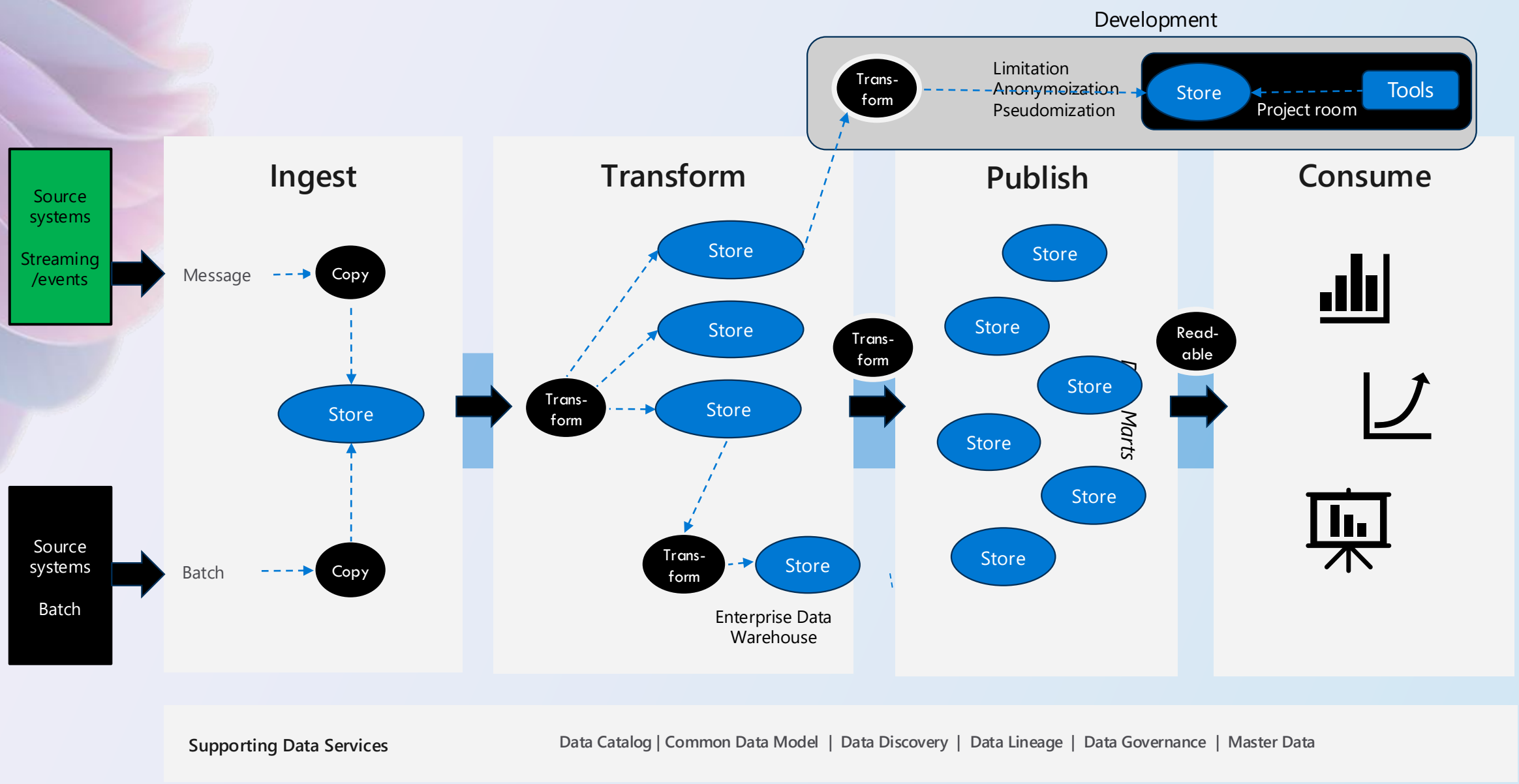
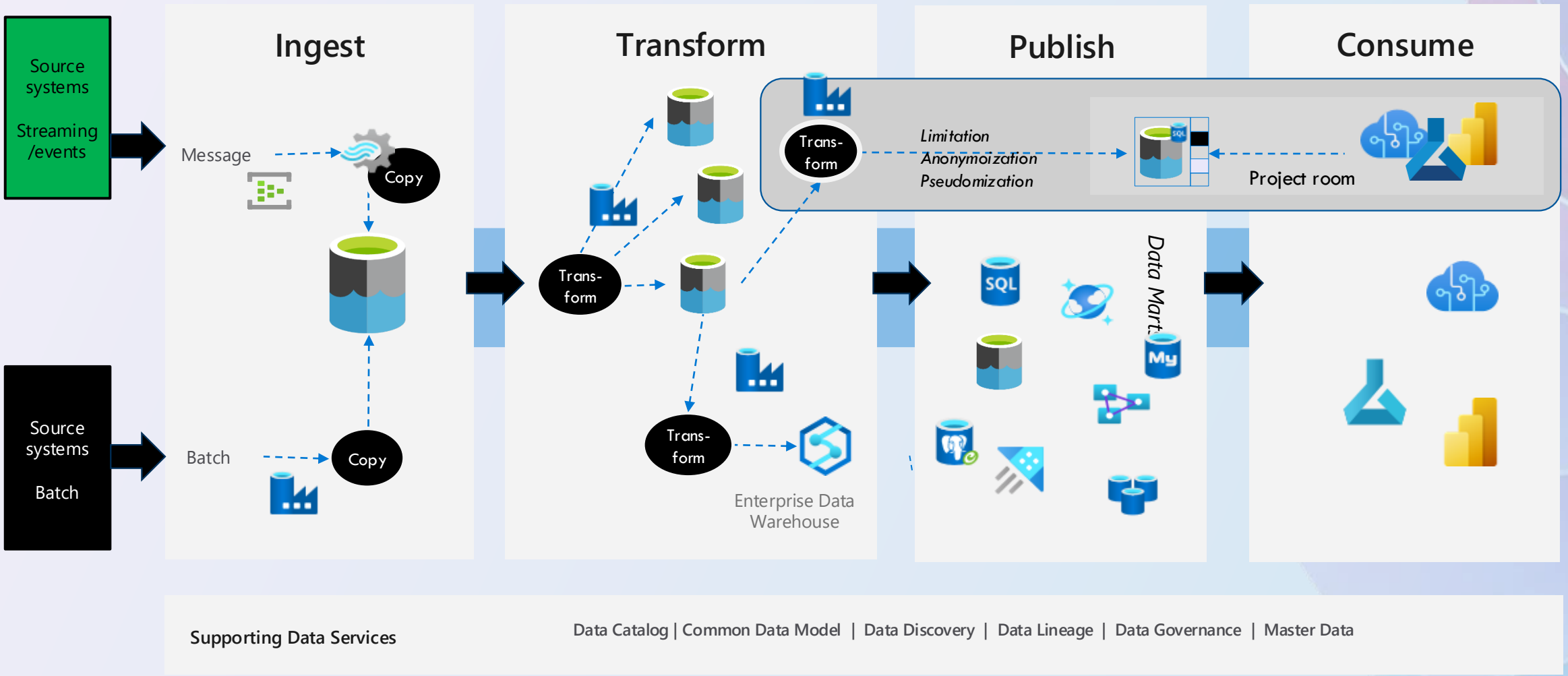
# How to……


Development
Trans-form
Limitation
Anonymoization
Pseudomization
Store
Tools
Project room
Source systems
Streaming /events
Source systems
Batch
Ingest
Message
Copy
Store
Batch
Copy
Transform
Trans-form
Store
Store
Store
Trans-form
Store
Enterprise Data Warehouse
Trans-form
Publish
Store
Store
Store
Store
Store
Store
Marts
Read-able
Consume
Supporting Data Services
Data Catalog | Common Data Model | Data Discovery | Data Lineage | Data Governance | Master Data

# Azure based




Supporting Data Services

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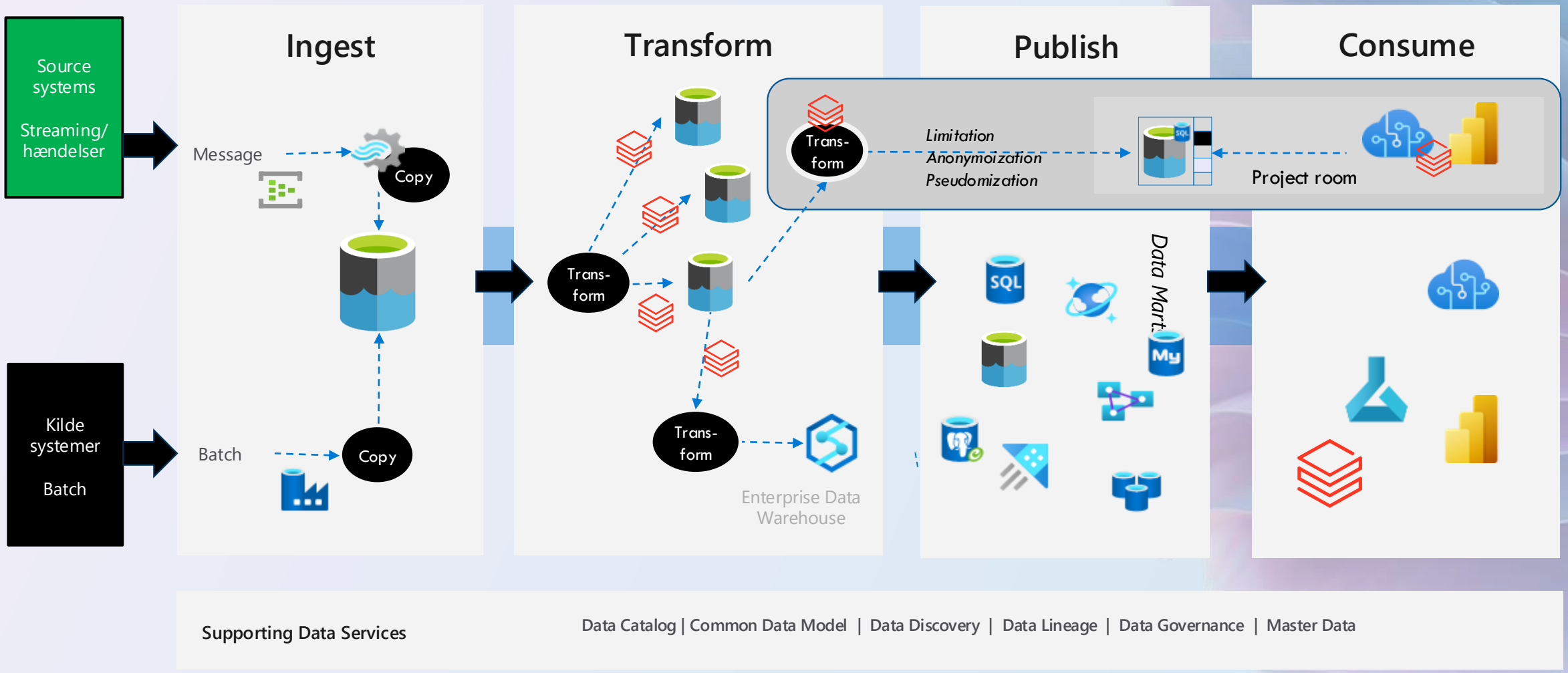
# Azure Databricks based




Supporting Data Services

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# Microsoft Fabric

## Intelligent data foundation

- Data Factory
- Data Engineering
- Data Warehouse
- Data Science
- Real Time Analytics
- Power BI
- Data Activator

Powered by AI with Copilot in Microsoft Fabric

**Single data foundation**

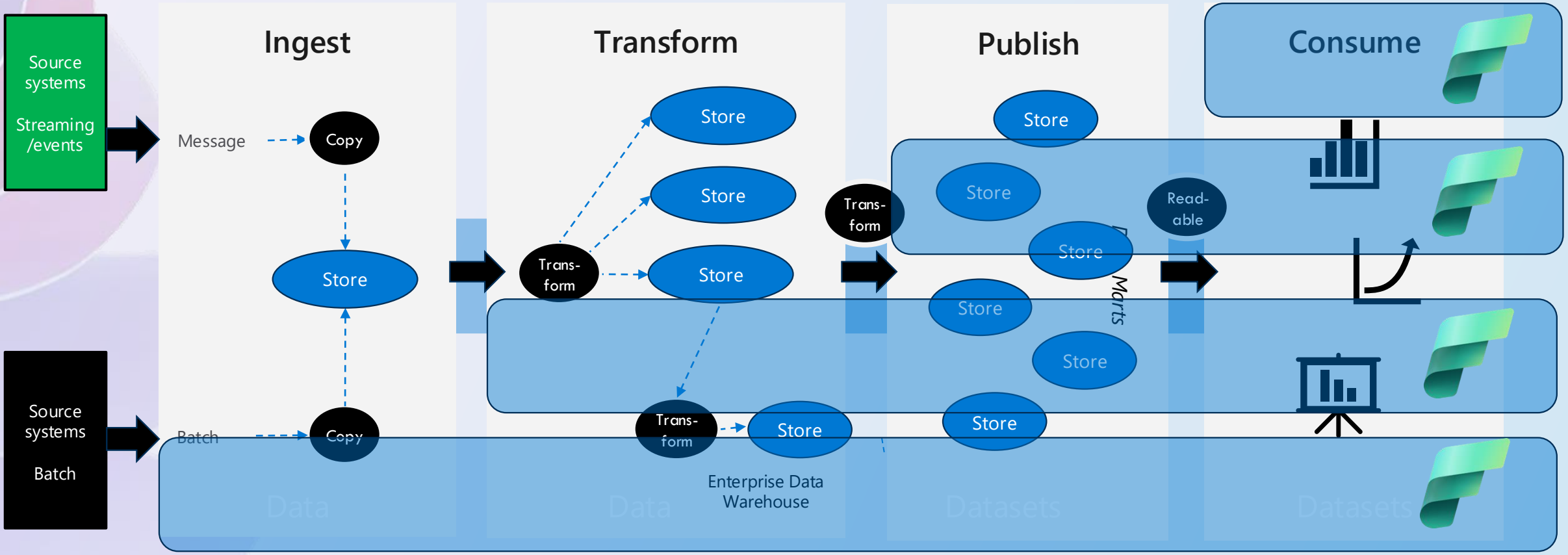
OneLake

**UNIFIED**

SaaS product experience | Security and governance | Compute and storage | Business model

Microsoft

# Fabric based




**Supporting Data Services** Data Catalog | Common Data Model | Data Discovery | Data Lineage | Data Governance | Master Data

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# Security

# Protection

## Role Based Access Control

Access control for accessing a resource

## Application Based Access Control

Access controlled through the application

Data Store

## Removal Based Access Control

If (when) “attacked” remove the service

## Attribute Based Access Control

Access control provided through a “lookup”

## Encryption Based Access Control

You need a key to read the data

## Identity Based Access Control

An application also have an identity

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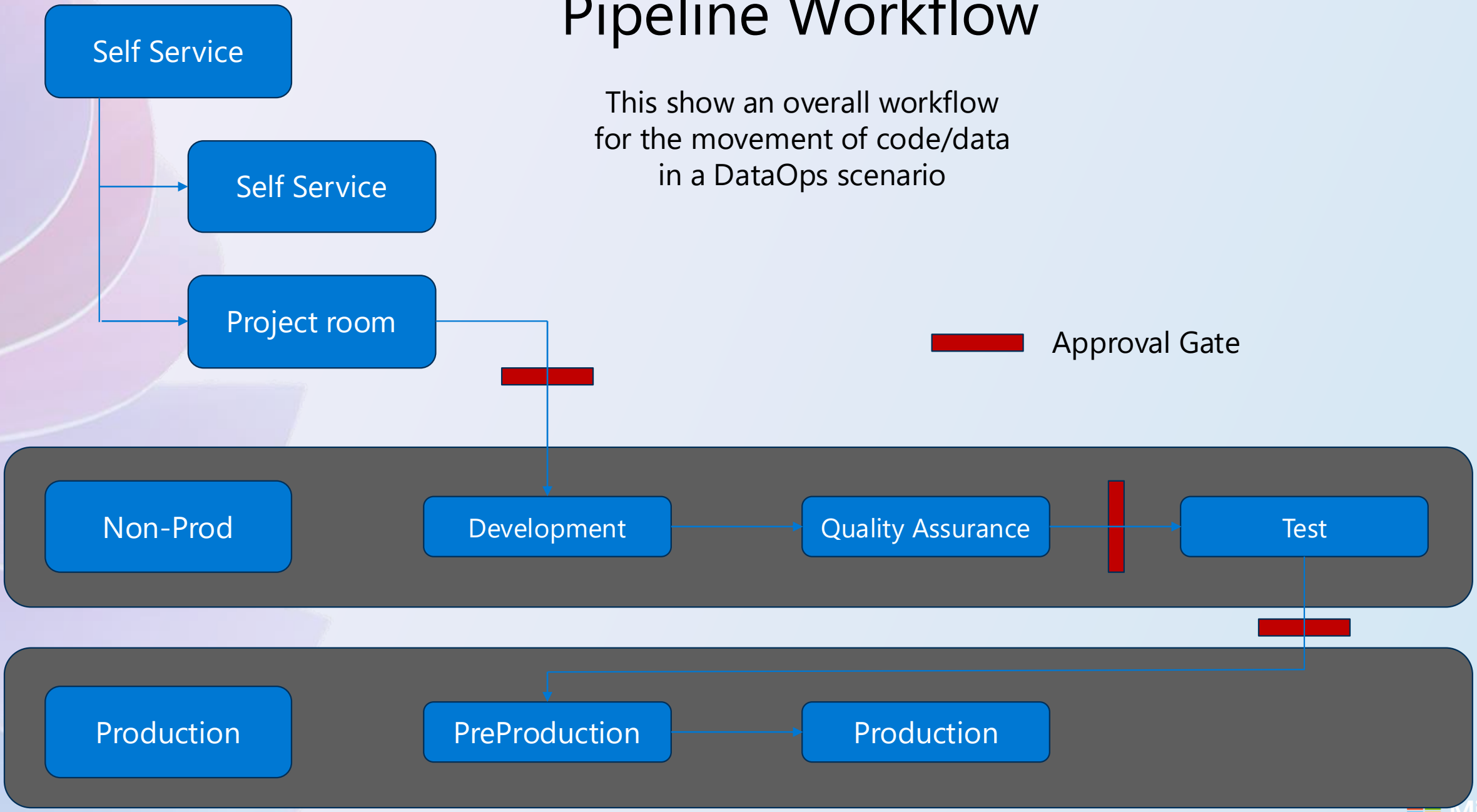
# Pipelining

# Pipeline Workflow

This show an overall workflow
for the movement of code/data
in a DataOps scenario

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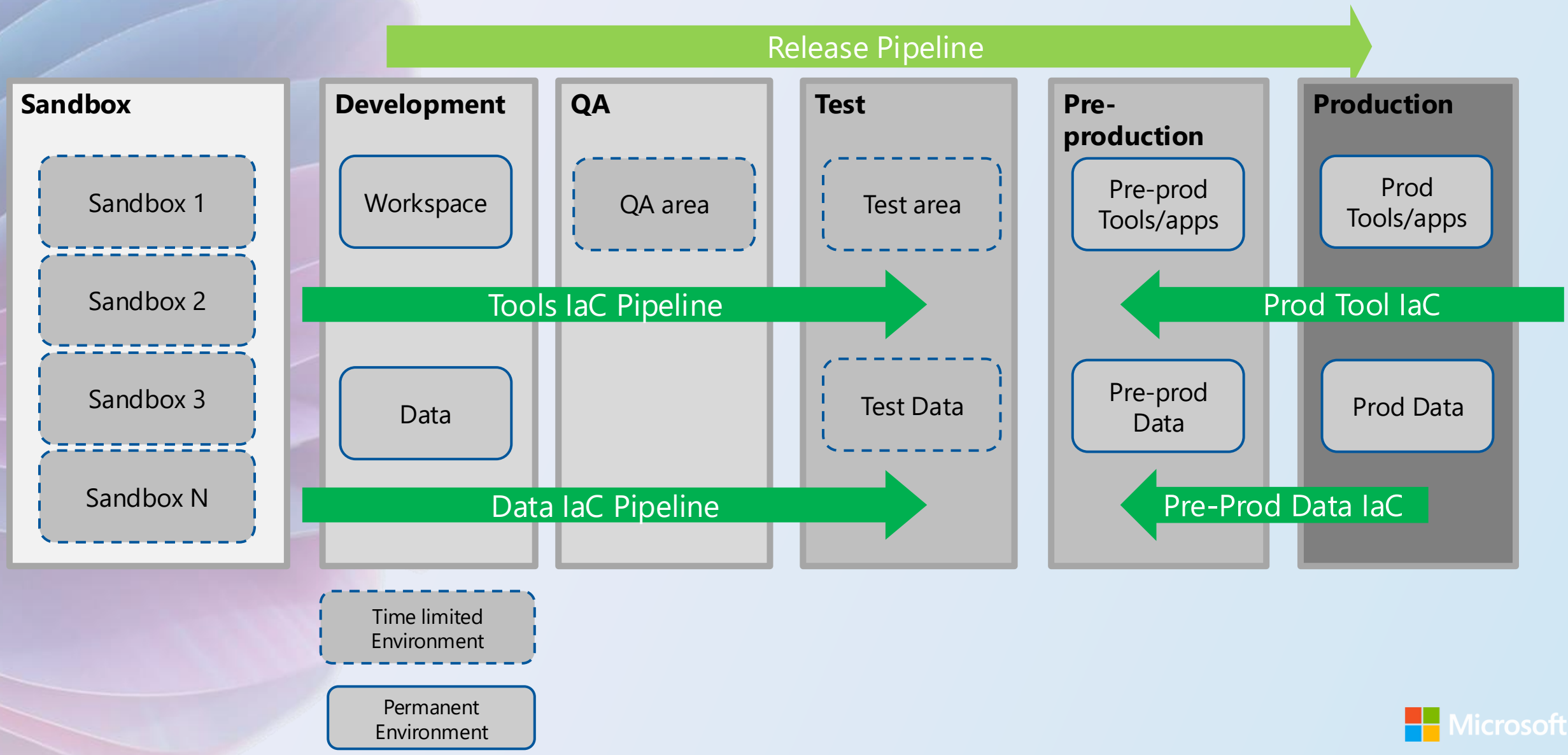
# Data : Environments / Pipelines

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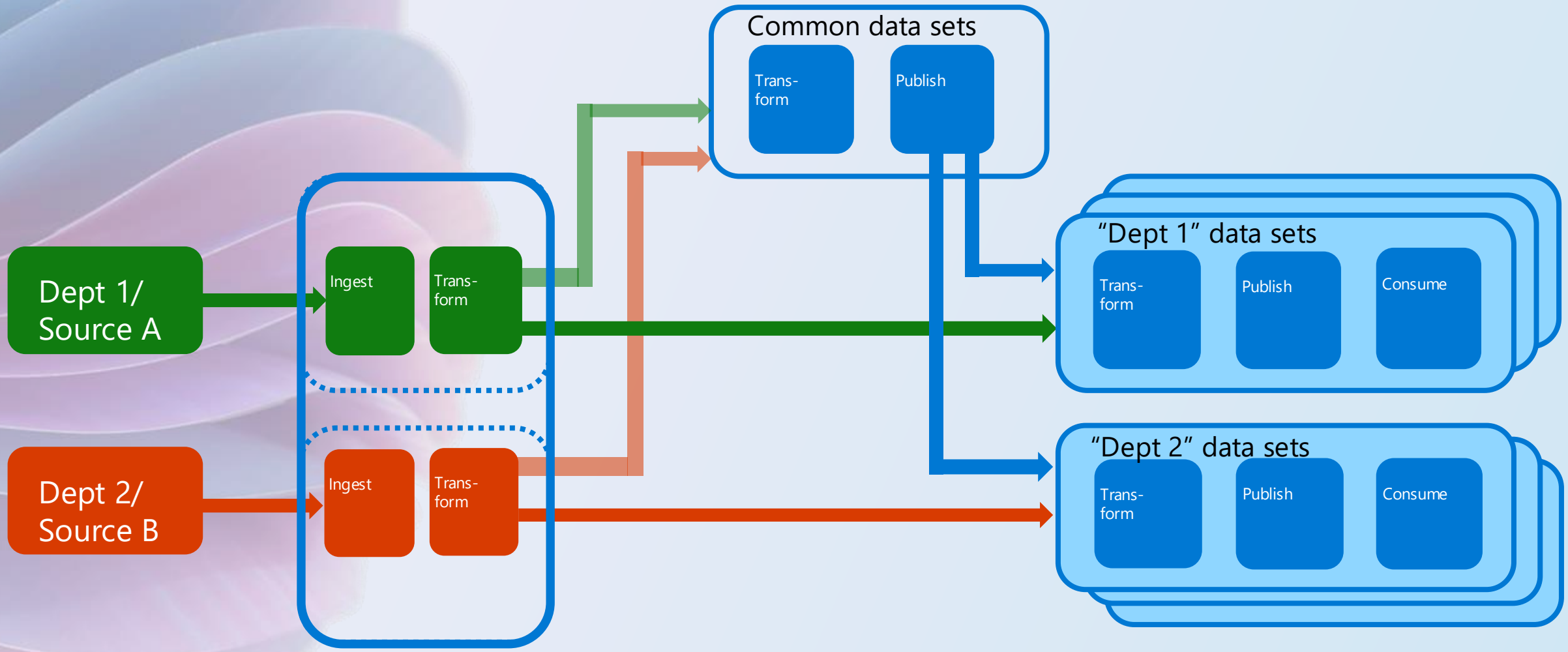
# Azure setup

# “Many data platforms”

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# Thank you

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https://github.com/jcordtz/a_data_platform

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